

Problem Set 2

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1 Natural Deduction

Problem 1: Prove the following statements using natural deduction. (The assumptions are to the right of the $\vdash$, the conclusions to the left.) Feel free to assume the *law of the excluded middle* $P \vee \neg P$ wherever appropriate.

1. $P \rightarrow Q, \neg Q \vdash \neg P$.
2. $P \wedge Q \vdash Q \vee R$.
3. $\vdash (P \rightarrow Q) \vee (Q \rightarrow P)$.

Problem 2: Prove the following statements using the Incredible Proof Machine <https://incredible.pm/>. You can just include a screencap of your result.

1. Session 1: $(A \wedge B \wedge C) \vdash A \wedge C$
2. Session 2: $A \rightarrow (B \rightarrow C) \vdash (A \wedge B) \rightarrow C$
3. Session 3: $(A \rightarrow C) \wedge (B \rightarrow C) \vdash (A \vee B) \rightarrow C$
4. Session 4: $(A \rightarrow \perp) \vee B \vdash (A \rightarrow B)$

2 Predicate Calculus

Problem 3: Formalize the following three statements in predicate calculus. (Remember to define your terms.)

1. You're not the only person. (You can use $x = y$ for identity.)
2. There is somebody who cares about everyone in the world. (Take this as literally referring to some individual, not in the sense of 4.2 below.)
3. There is someone who loves all, and only, people who don't love themselves.

Problem 4: Now use the above claims to prove the following (respective) conclusions:

1. There exists a person.
2. Everyone has somebody who cares about them.
3. **Bonus** False. (Hint: Think about Russell's Paradox [lecture 5, also on Wikipedia] and use the law of the excluded middle.)

Problem 5: Prove the following statements using the Incredible Proof Machine. You can just include a screencap of your result:

1. Session 6: One of the first six problems.
2. Session 6: Any other problem.